VIT BHOPAL

"Cafe Management System - Running Technology"

Cafe Management System

Fundamentals of AI and ML

Registration no.:
25BCE11035

Faculty:
M.K.Jayanthi

TRUSHA SHIVAJI DHABE

Introduction

This project is focused on the creation of a Smart Cafe Management System (CAFEM) using the Python programming language. The intention of this project is to enhance various aspects of daily café operation through improved management functions, enhanced feature sets, intelligent feedback suggestions, and well-organized development practices. By demonstrating the ability of using software application to create a more efficient operational process for small businesses, the project aims to present the benefits of improved electronic processing to enhance small café operations through the application of electronic technology in a simple console-based application environment.

Motivation of the Project

Small businesses, such as cafes, operate in an increasingly fast-paced atmosphere and therefore need to manage systems efficiently if they are going to stay organized and continue delivering competitive service. Unfortunately, the majority of cafes still use manual systems that create high levels of error and provide slow service with an unacceptable overall customer experience. This project aims to simplify and assist the operation

of cafes through the development of intelligent features, such as detailed recommendations and insights about customers in relation to the cafe's business model. The principal aim of the project is to develop a simplistic, yet intelligent, single, and easy-to-use management system for cafes to incorporate into their operations. The project will use Python to write about the management system and integrate intelligent features.

Problem Statement

When you run a cafe, you have many responsibilities such as managing your menu, taking customer orders, tracking sales, and analyzing customer behavior. Some of these responsibilities can be done manually or without structure. This often causes problems like confusion, slow service, and errors. This project will address these problems with a structured and automated solution that will improve the efficiency of your business while enhancing the overall experience for customers.

Objective of the Project

- To develop a system that manages café operations efficiently
- To reduce manual effort and human errors
- To provide basic business insights through sales analysis

- To enhance customer experience using smart recommendations
- To demonstrate real-world application development using Python

Existing Methods

- Manual record keeping using notebooks
- Basic billing systems without analytics
- Simple POS systems with limited features

Pros and Cons of Existing Methods

Pros:

- Easy to start with
- No technical knowledge required
- Low initial cost

Cons:

- Time-consuming and error-prone
- No proper data tracking or insights
- Difficult to manage as business grows
- No personalization for customers

Hardware and Software Requirements

Hardware:

- Basic computer or laptop
- Minimum 4GB RAM

Software:

- Python (3.x)
- Any code editor (VS Code, PyCharm, etc.)
- Operating System: Windows/Linux/Mac

Methodology and Goal

The project is organized into several different components including menu management, order processing and reporting, and uses an organized approach in the development of each of those components.

An easy-to-use system should still include advanced features, such as modular design, recommendation logic, and data handling.

Functional Modules Design and Analysis

- **Menu Management Module**
  Handles adding, updating, viewing, and deleting menu items
- **Order Processing Module**
  Manages order placement, validation, and billing
- **Recommendation Module**
  Suggests items based on popularity or user behavior
- **Customer Classification Module**
  Categorizes users into VIP, Regular, etc.
- **Sales Analysis Module**
  Displays sales trends and performance

Algorithm Development

Example: Recommendation Logic

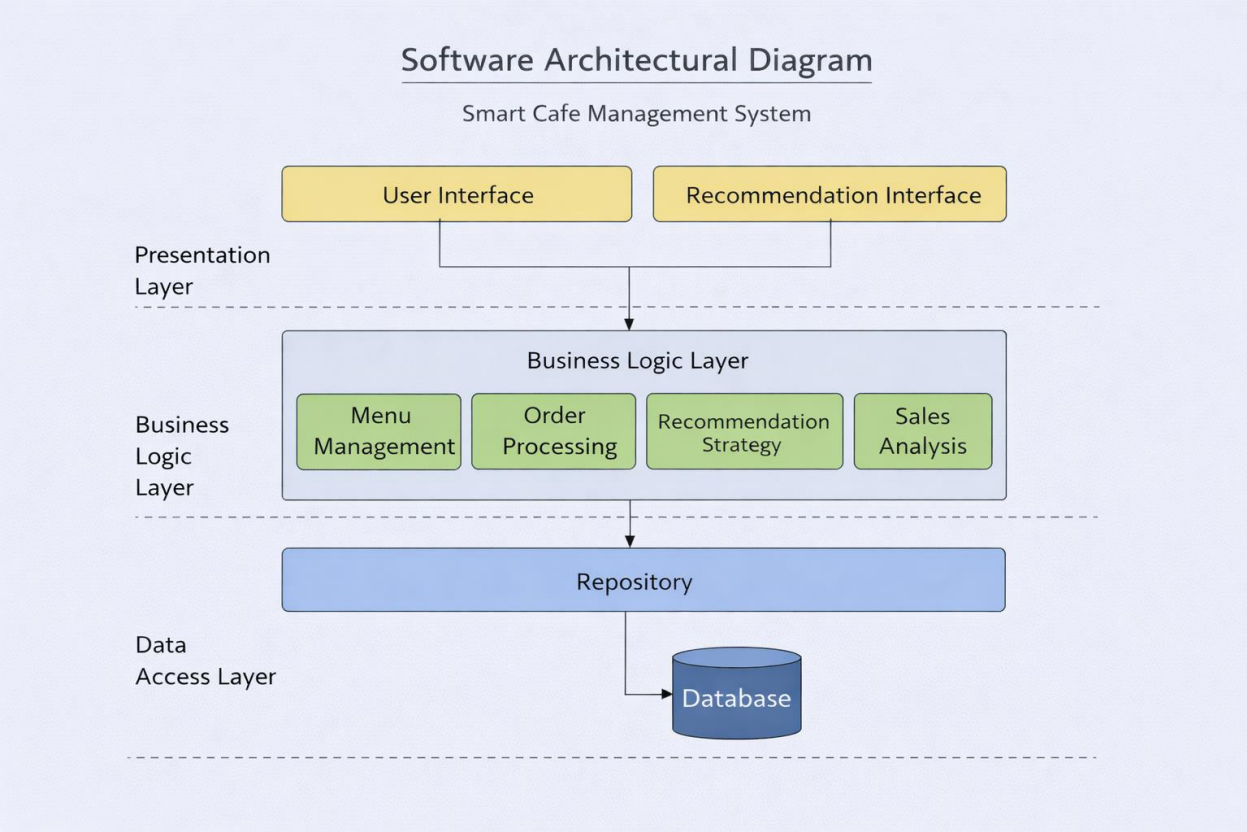
1. Collect order history
2. Identify frequently ordered items
3. Rank items based on popularity
4. Suggest top items to the user

Example: Order Processing

1. User selects items

2. Validate input
3. Calculate total price
4. Generate receipt

Software Architectural Diagram



Coding

The Smart Cafe Management System is built using Python. It is designed in an organized way. The Smart Cafe Management System has parts that work together. Each part of the Smart Cafe Management System does a job, like handling menus processing orders for the Smart Cafe Management System and making

reports, for the Smart Cafe Management System. This makes the Smart Cafe Management System easy to understand and work with. The Smart Cafe Management System is also easy to maintain and add things to the Smart Cafe Management System.

Output

The system provides:

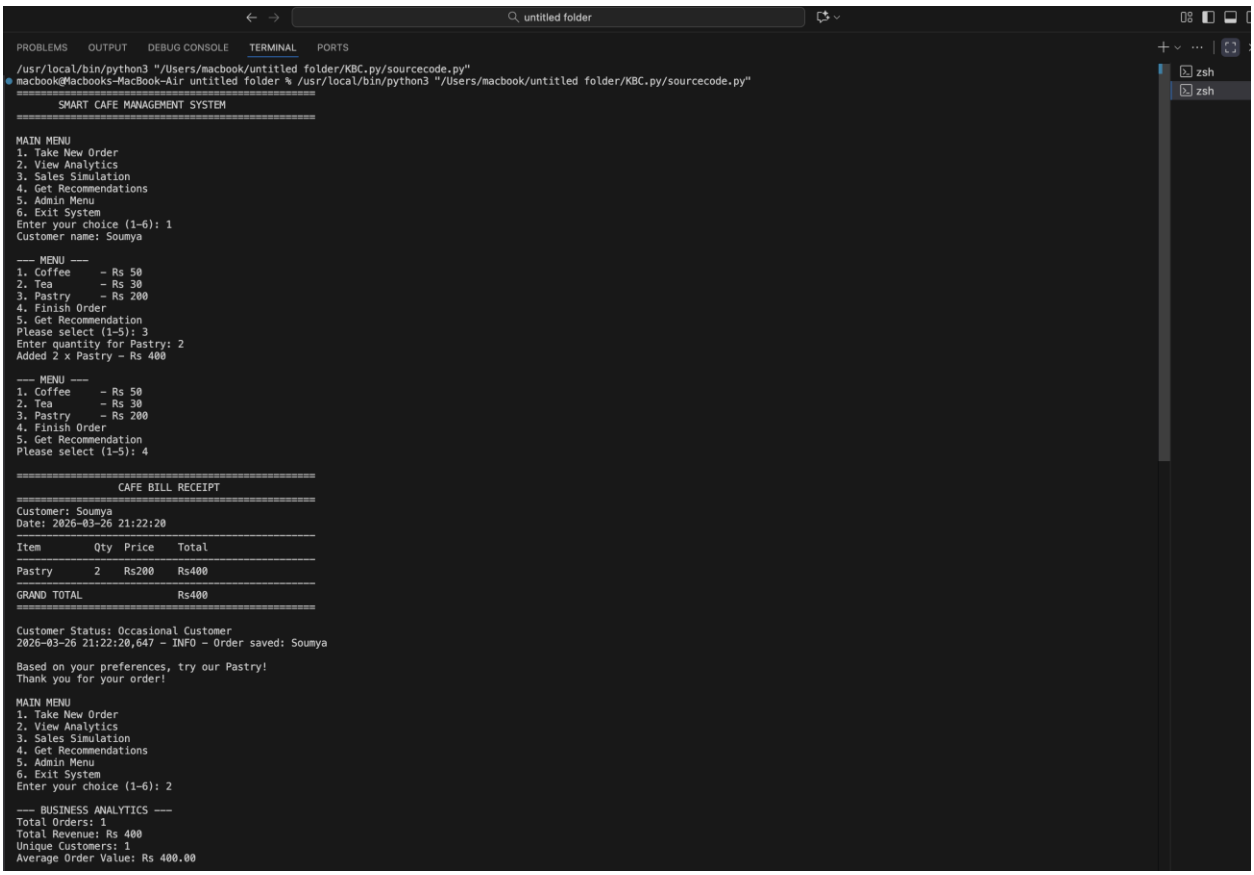
- Menu display
- Order confirmation
- Generated receipt
- Sales reports

Key Implementations Outline of the System

- CRUD operations for menu
- Recommendation system
- Customer classification logic
- Sales simulation and reporting
- Clean modular code structure

Significant Project Outcomes

- Improved understanding of real-world systems
- Successfully implemented multiple modules
- Demonstrated use of design patterns
- Built a working end-to-end system



```
untitled folder
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
/usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"
macbook@Macbooks-MacBook-Air untitled folder % /usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"
==================================================
        SMART CAFE MANAGEMENT SYSTEM
==================================================

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 1
Customer name: Soumya

--- MENU ---
1. Coffee     - Rs 50
2. Tea        - Rs 30
3. Pastry     - Rs 200
4. Finish Order
5. Get Recommendation
Please select (1-5): 3
Enter quantity for Pastry: 2
Added 2 x Pastry - Rs 400

--- MENU ---
1. Coffee     - Rs 50
2. Tea        - Rs 30
3. Pastry     - Rs 200
4. Finish Order
5. Get Recommendation
Please select (1-5): 4

==================================================
                CAFE BILL RECEIPT
==================================================
Customer: Soumya
Date: 2026-03-26 21:22:20
--------------------------------------------------
Item         Qty  Price    Total
--------------------------------------------------
Pastry        2   Rs200    Rs400
--------------------------------------------------
GRAND TOTAL                Rs400
==================================================

Customer Status: Occasional Customer
2026-03-26 21:22:20,647 - INFO - Order saved: Soumya

Based on your preferences, try our Pastry!
Thank you for your order!

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 2

--- BUSINESS ANALYTICS ---
Total Orders: 1
Total Revenue: Rs 400
Unique Customers: 1
Average Order Value: Rs 400.00
```

```
macbook@Macbooks-MacBook-Air untitled folder % /usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 2

--- BUSINESS ANALYTICS ---
Total Orders: 1
Total Revenue: Rs 400
Unique Customers: 1
Average Order Value: Rs 400.00

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 3

--- 7-DAY SALES SIMULATION ---
Day 1: Rs 3193
Day 2: Rs 4943
Day 3: Rs 2196
Day 4: Rs 4446
Day 5: Rs 3517
Day 6: Rs 6275
Day 7: Rs 4197

Analytics:
   Average: Rs 4110
   Best: Rs 6275
   Worst: Rs 2196
   Total: Rs 28767

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 4
Enter customer name: Trusha

Recommendation for Trusha:
Popular choice: Pastry!

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 5
```

```
macbook@Macbooks-MacBook-Air untitled folder % /usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"
Day 1: Rs 3193
Day 2: Rs 4943
Day 3: Rs 2196
Day 4: Rs 4446
Day 5: Rs 3517
Day 6: Rs 6275
Day 7: Rs 4197

Analytics:
   Average: Rs 4110
   Best: Rs 6275
   Worst: Rs 2196
   Total: Rs 28767

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 4
Enter customer name: Trusha

Recommendation for Trusha:
Popular choice: Pastry!

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 5

--- ADMIN MENU ---
1. Add Item
2. View All Items
3. Update Item
4. Delete Item
5. Back to Main Menu
Choice: 3
Item ID to update: 4
Error: Item not found!

--- ADMIN MENU ---
1. Add Item
2. View All Items
3. Update Item
4. Delete Item
5. Back to Main Menu
Choice: 5

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 6

==================================================
            CAFE SYSTEM CLOSED
==================================================
Final Report: 1 orders, Rs 400 revenue
Thank you!
macbook@Macbooks-MacBook-Air untitled folder %
```

Testing and Refinement

The system is tested using different input scenarios to ensure it behaves correctly. This includes:

- Valid and invalid user inputs
- Edge cases like empty orders or unavailable items
- Repeated operations to check stability

Project Applicability on Real-World Applications

This system can be used as a base for real café software. With further development, it can be extended into:

- Restaurant management systems
- Online food ordering platforms
- POS systems

Contribution / Findings of the Project

- Shows how automation improves efficiency
- Demonstrates importance of structured coding
- Highlight's role of data in decision-making

Limitations / Constraints of the System

- No graphical interface (console-based only)
- No real database integration
- Limited scalability
- Recommendation system is basic

Conclusions

This project has shown through its use of technology in managing cafes, that technology such as the Smart Cafe Management Program can effectively reduce the burden of operating a cafe while providing enhanced efficiencies. It also highlights the benefits of automating processes, being organized, and designing products with the end user in mind. This project is an excellent base to develop more complex systems in the real world.

Future Enhancements

- Add a graphical user interface (GUI)
- Integrate a real database for persistent storage
- Include online ordering and payment features
- Improve recommendation system using advanced algorithms

References

- Python official documentation
- Online tutorials and learning resources
- Basic concepts of software design patterns